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Abstract

Background

- Meibomian gland dysfunction (MGD) is a major cause of evaporative dry eye disease (DED).
- Aging is a key risk factor for MG degeneration.
- Effective treatments are limited due to a lack of clearly defined therapeutic targets.

Purpose

- To determine the function of Hedgehog (Hh) signaling on MG development and adult maintenance
- To identify downstream targets that regulate MG development and may be implicated in age-related MGD.

Materials & Methods

- Transcriptomic analysis of age-related changes in Hh pathway activity.
- K14-Cre-based Smo and Ifi88 ablation models in MG epithelium.
- *Gli1-LacZ* and *Gli1-CreERT2*; *Rosa26mTmG* reporter mice to detect and trace Hh-responsive cells.
- RT-qPCR, RNAscope, immunostaining, EdU, Oil Red O staining, and lipidomic analysis to assess gene expression, proliferation, MG morphology, and lipid homeostasis.

Results

- Hh pathway activity declines with age in human tarsal plate and decreases during postnatal MG maturation in mice.
- *Smo* deletion impairs MG growth, morphology, lipid accumulation, and lipid composition.
- *Iff88* ablation partially rescues *Smo*-dependent MG defects and restores lipid homeostasis.
- Double mutants show partial recovery of Hh target genes, including *Gli1* and *Ptch1*.
- Hh signaling supports epithelial proliferation and ductal progenitor programs, and *Gli1*+ ductal cells contribute to the adult duct–acinar lineage.

Conclusions

- Hh signaling maintains MG progenitors, epithelial growth, and lipid homeostasis.
- Targeting the cilia–Hh axis may help restore MG function in age-related MGD and evaporative DED.

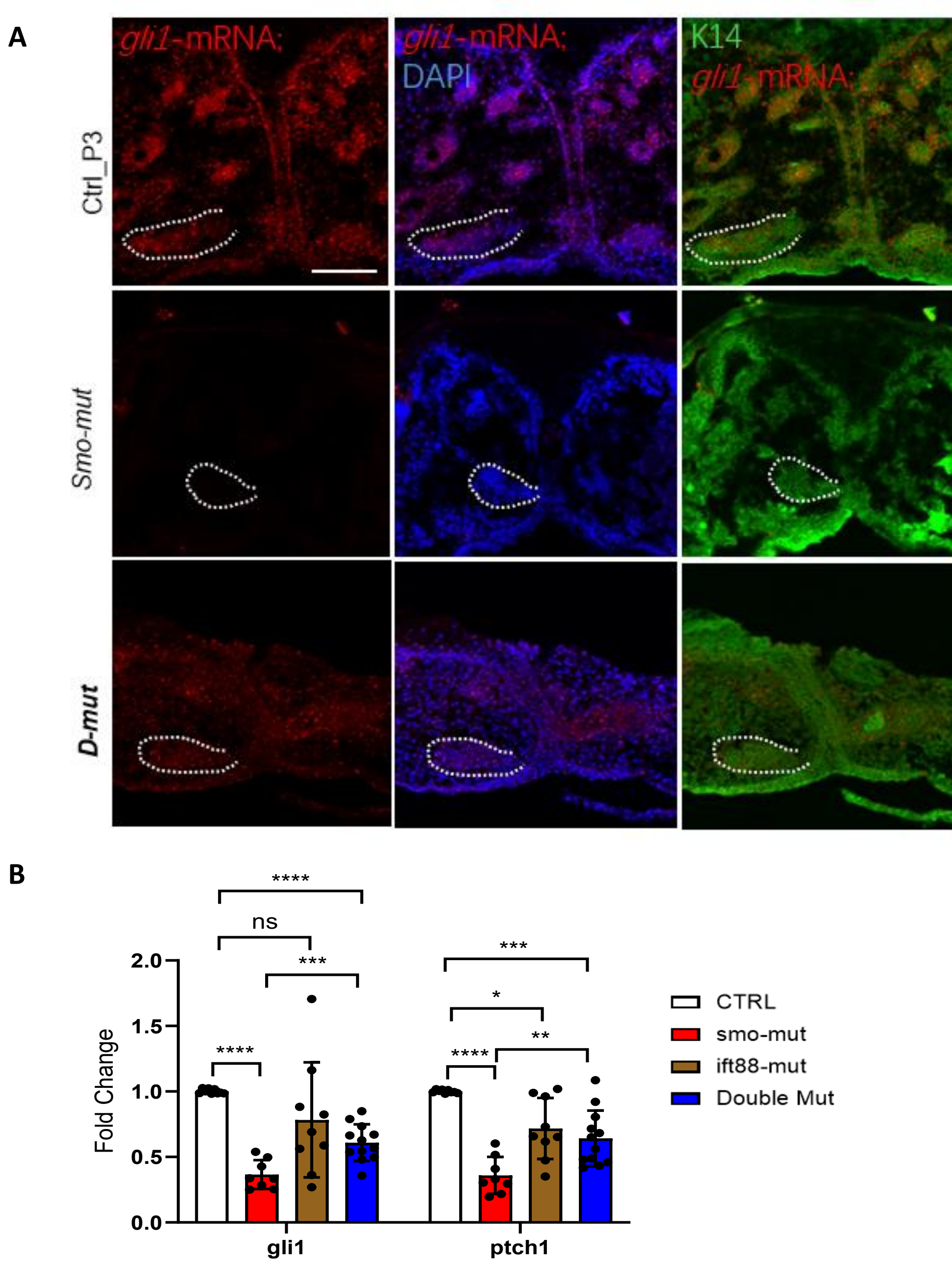


Figure 6. Hh target gene expression is partially restored in double mutants. A) RNAscope staining shows reduced Gli1 mRNA expression in Smo mutant meibomian glands and partial restoration in double mutants at P3. Dashed outlines indicate the developing meibomian gland region. B) qPCR analysis shows partial recovery of Hedgehog target genes Gli1 and Ptch1 in double mutants. scale bar, 100 μ m.

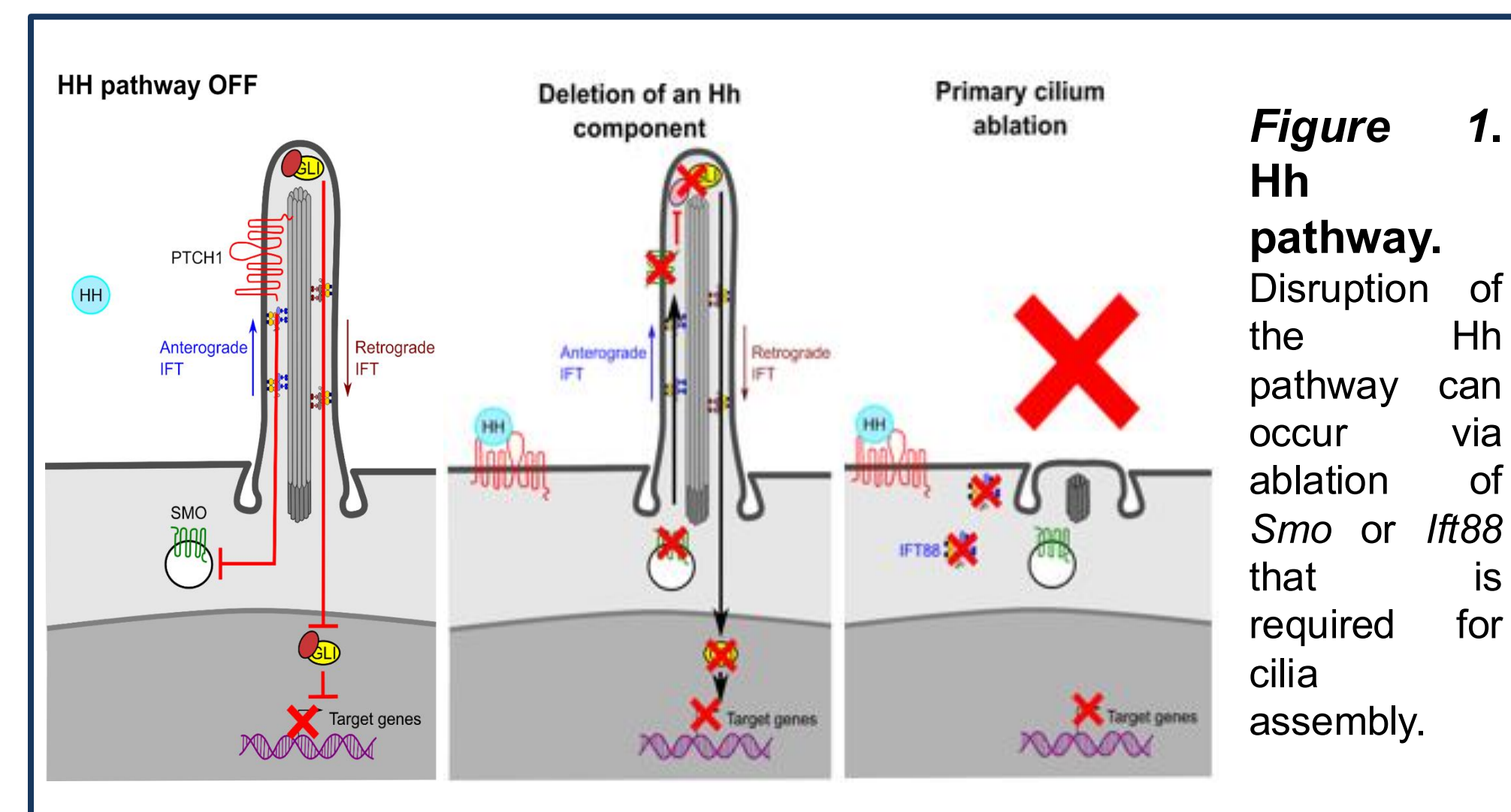


Figure 1.
Hh pathway.
Disruption of the Hh pathway can occur via ablation of *Smo* or *lft88* that is required for cilia assembly.

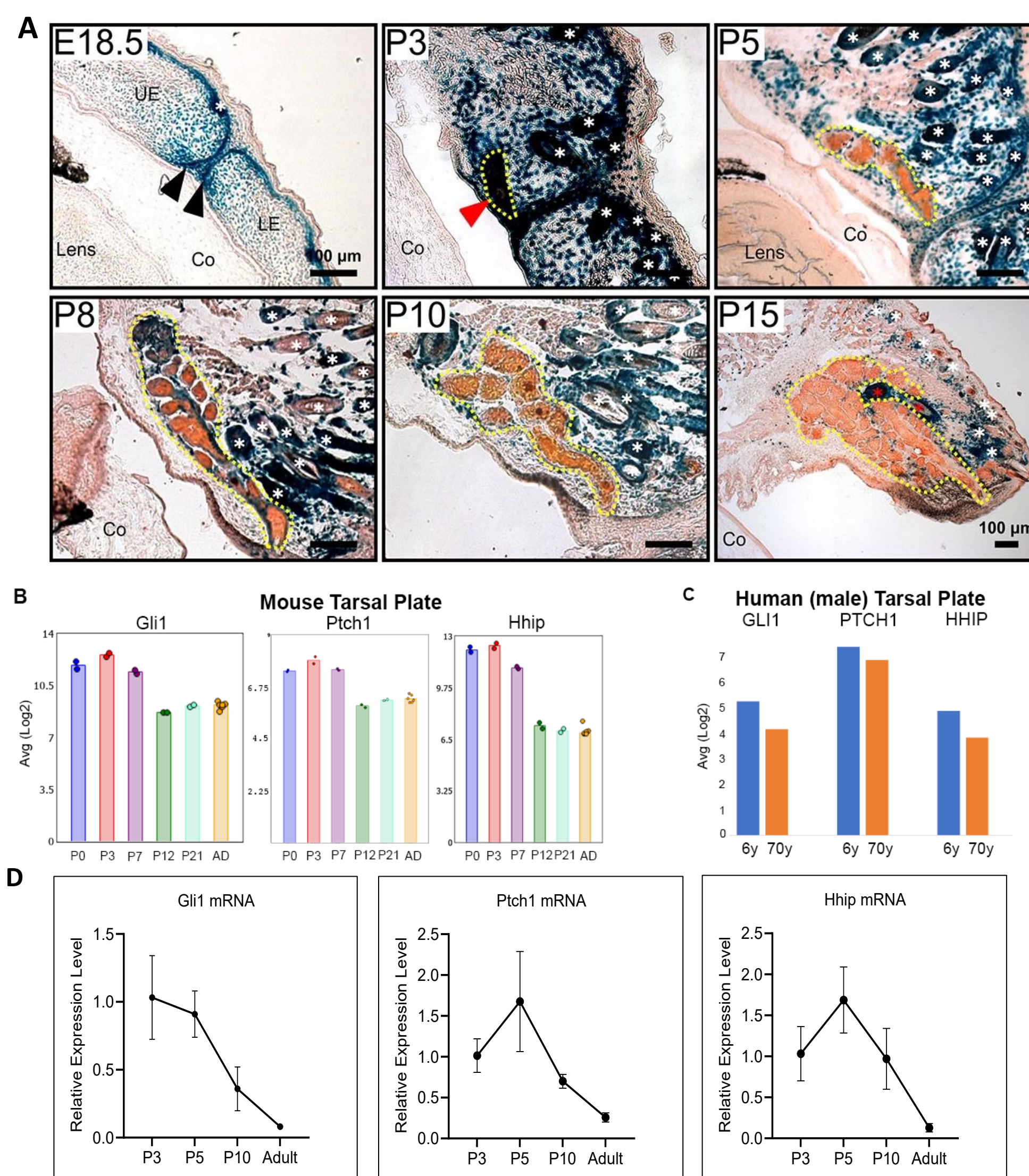


Figure 2. Hh activity decreases as postnatal development progresses in mouse & human tarsal plates(TPs). A) Gli1- β -galactosidase expression in the mouse TPs. At E18.5 Hh activity, seen as blue staining, is present at the fused eyelid margin where MG begins to develop (black arrows). At P3, Hh is highly activated in the central duct. By P5, MG appears as clusters of orange cells due to lipid staining with ORO. Between P5 and P10, Hh responsiveness decreases as the MG branches. By P15, Hh responsive cells are restricted to discrete areas of the acini and in the excretory duct (yellow). B) and C) Expression profile of Hh target genes in the TPs of B) Mouse and C) Human at different postnatal stages from an unbiased transcriptomic analysis using HTA-2.0 mRNA microarray. D) qPCR verified how a postnatal decline of Gli1, Ptc1, and Hhip expression in mouse TPs from P3, P5, P10 to adult stages.

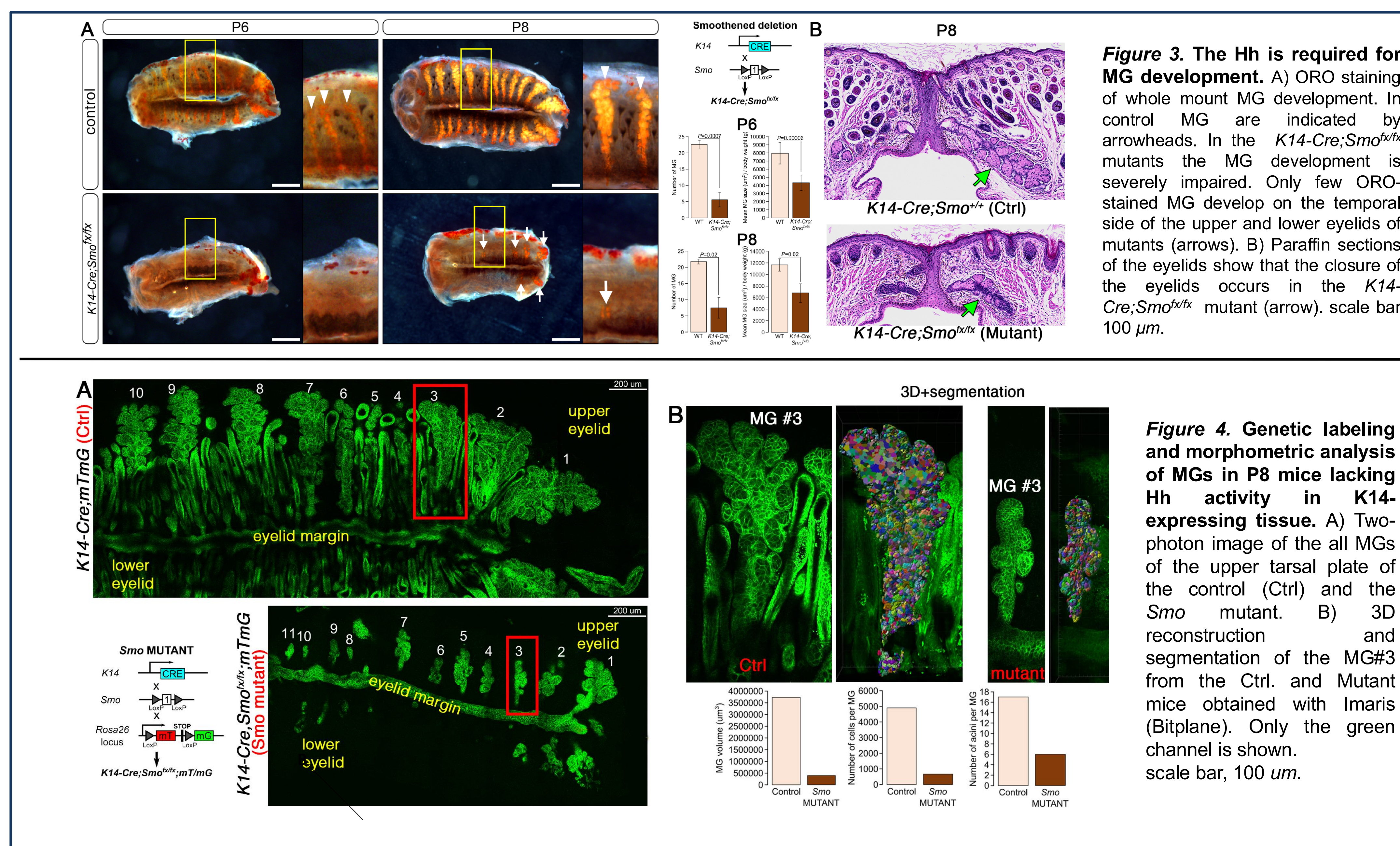


Figure 3. The Hh is required for MG development. A) ORO staining of whole mount MG development. In control MG are indicated by arrowheads. In the *K14-Cre;Smo^{Δf/tx}* mutants the MG development is severely impaired. Only few ORO-stained MG develop on the temporal side of the upper and lower eyelids of mutants (arrows). B) Paraffin sections of the eyelids show that the closure of the eyelids occurs in the *K14-Cre;Smo^{Δf/tx}* mutant (arrow). scale bar 100 μ m.

Figure 4. Genetic labeling and morphometric analysis of MGs in P8 mice lacking Hh activity in K14-expressing tissue. A) Two-photon image of the all MGs of the upper tarsal plate of the control (Ctrl) and the *Smo* mutant. B) 3D reconstruction and segmentation of the MG#3 from the Ctrl. and Mutant mice obtained with Imapris (Bitplane). Only the green channel is shown. scale bar, 100 μ m.

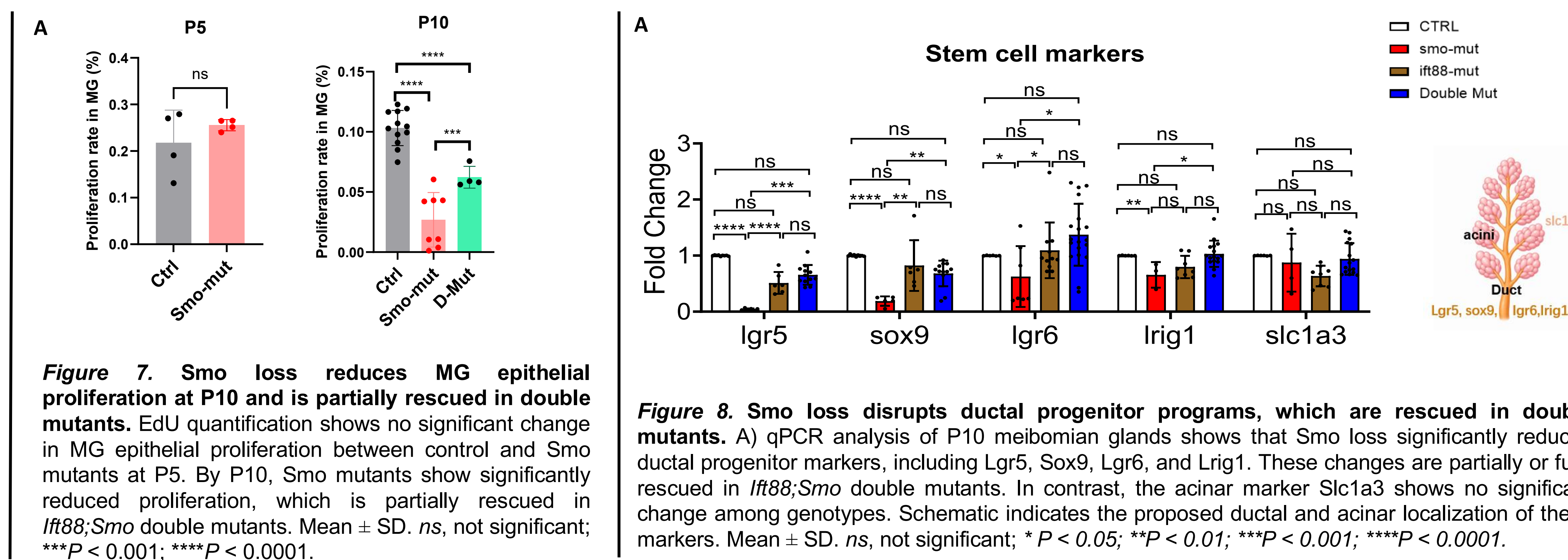


Figure 8. Smo loss disrupts ductal progenitor programs, which are rescued in double mutants. A) qPCR analysis of P10 meibomian glands shows that Smo loss significantly reduces ductal progenitor markers, including *Lgr5*, *Sox9*, *Lgr6*, and *Lrig1*. These changes are partially or fully rescued in *lft88;Smo* double mutants. In contrast, the acinar marker *Slc1a3* shows no significant change among genotypes. Schematic indicates the proposed ductal and acinar localization of the markers. Mean $\pm$ SD. ns, not significant; * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; **** $P < 0.0001$.

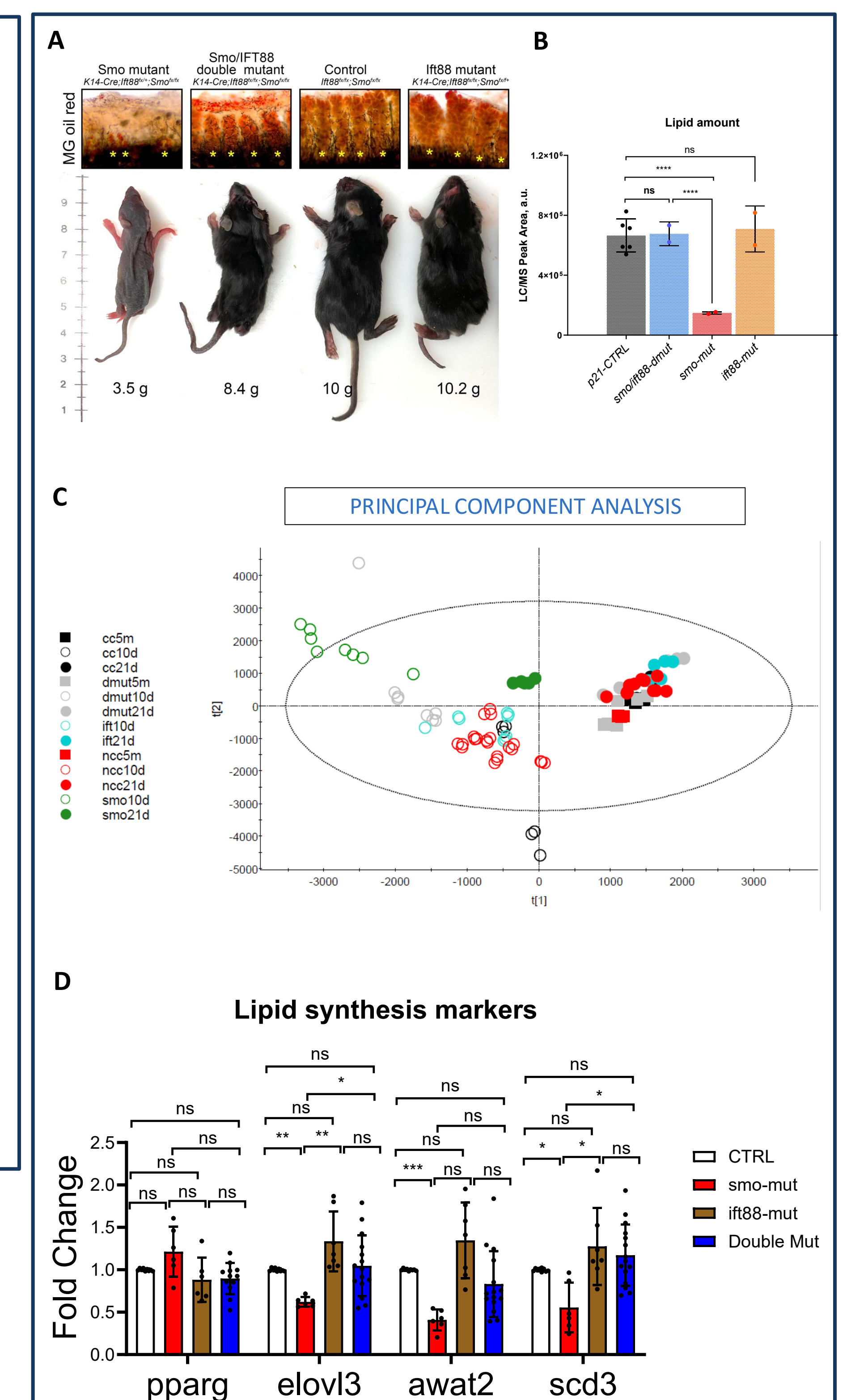


Figure 5. Loss of cilia partially rescues Smo-dependent MG defects and restores lipid homeostasis.

A) Genetic ablation of *lft88* in *K14+* tissue partially rescues the atrophic meibomian gland (MG) phenotype caused by *Smo* loss. Oil *Red O* staining shows improved MG morphology and lipid accumulation in *lft88*; *Smo* double mutants compared with *Smo* mutants. Lipidomic analysis further shows restoration of total lipid levels B) and overall lipid composition C), including major meibum lipid families, in double mutants. D) qPCR analysis of lipid synthesis markers support a partial recovery of lipid homeostasis.

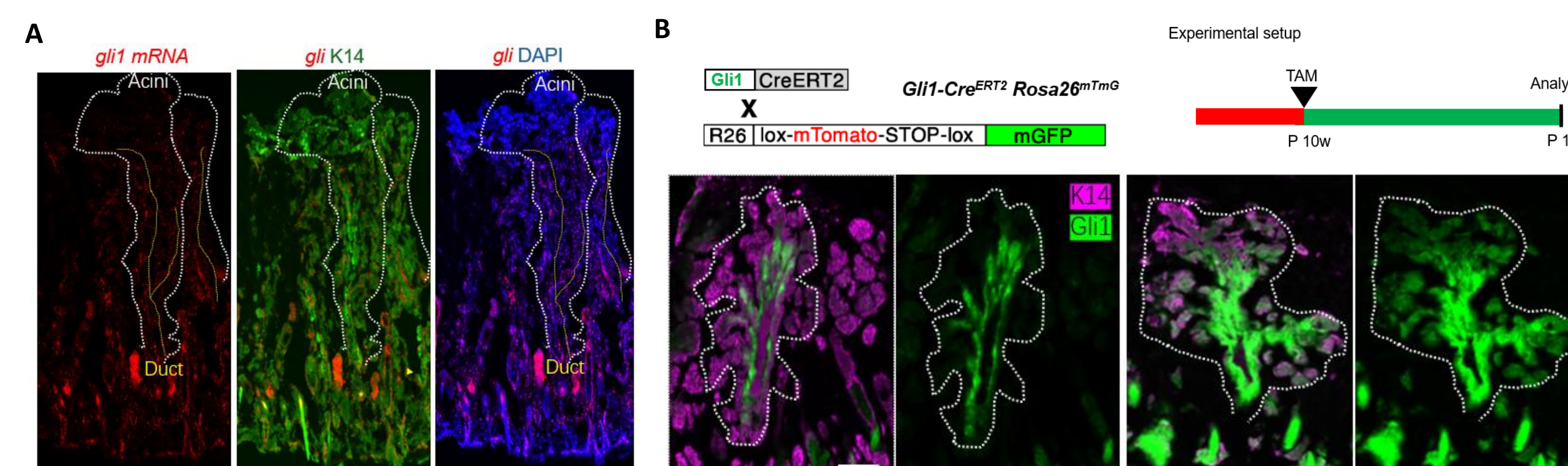


Figure 9. Hh-responsive ductal cells contribute to the duct–acinar lineage in adult MG. A) *Gli1* RNAscope identifies Hh-responsive cells enriched in the adult MG duct, and B) *Gli1*-lineage tracing shows their contribution to the duct–acinar axis over time. Dashed outlines indicate MG structures. Scale bar, 100 μ m.

Working models

- Hh signaling is essential for MG development and homeostasis
- Hh maintains ductal stem/progenitor activity in the MG
- Targeting cilia–Hh signaling may offer a therapeutic strategy for age-related dry eye disease

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